

UNIT 1: THINKING ABOUT OUR FOOTPRINT

Class Time: 75 Minutes

Description

This lesson will encourage the learner to think critically about their actions and evaluate their roles and responsibilities in environmental sustainability. The lesson will begin with a brief illustration of the interdependence of natural phenomenon and the fragile complexity of the planet's life-support systems. This activity is meant to draw the student into the vastness of the concept of sustainable development and demonstrate how our values influence our perceptions on the environment (See Activity 1: Yarn of Life). As a homework exercise, students will take a close look at their values and determine whether or not they put these values into practice. To do this, students are asked to bring a copy of the ecological footprint quiz home to determine the impact of their actions on the earth (See Activity 2: The Ecological Footprint).

Drawing from the values identified in the first activity, students then explore the issues of sustainable development according to what they think are the most important issues for Canada. Students must first reach consensus in small groups on their list of priorities and then develop one set of priorities for the entire class. Following this, students must determine who has the most responsibility for ensuring that these priorities are achieved (See Activity 3: Dismantling Sustainable Development).

At this point, students will be given a group assignment to examine the historical path of sustainable development. Students will be able to identify the policies and international commitments that coincide with the priorities they identified as a class. Students will be asked to discuss their findings in the third unit of this module (See Activity 4: The Historical Path of Sustainable Development).

Curriculum Expectations

The Yarn of Life

Overall Expectations

HEV.03 – evaluate approaches, policies, and principles relating to the protection and sustainability of the planet's life-support systems.

Specific Expectations

HE3.01 – explain the interactive nature of selected natural and human systems.

GC1.01 – identify current global sustainability issues and environmental threats.

The Ecological Footprint: Homework

Specific Expectations

UC2.02 - evaluate the sustainability of selected trends related to consumption of the earth's resources.

Dismantling Sustainable Development

Overall Expectations

HEV.03 – analyse geographic issues that arise from the impact of human activities on the environment in different regions of the world.

Specific Expectations

GC1.01 – identify current global issues and environmental threats.

HE1.03 – explain how people perceive resources and sustainable development differently at different times and in different places.

UC2.03 – evaluate the role played by non-governmental organisations and local community initiatives in different parts of the world in promoting sustainable development and resource management.

The Sustainable Development Timeline: Homework

Overall Expectations:

SSV.03 – demonstrate an understanding of the cultural, economic, and political aspirations of selected groups and the effects of their actions on local, national, and global issues.

HEV.03 – evaluate approaches, policies, and principles relating to the protection and sustainability of the planet's life-support systems.

GCV.03 – identify the contributions made by a variety of individuals, organizations, and institutions to sustainable development strategies for the developing world, and evaluate their economic, environmental and social impact.

UCV.03 – evaluate the effectiveness of methods used by different organizations, governments, and industries to find short- and long-term solutions to geographic problems and issues at the local, national and global level.

Specific Expectations:

SS1.05 – identify ways in which countries and regions are becoming increasingly interdependent.

HE1.03 – explain how people perceive resources and sustainable development differently at different times and in different places.

HE2.05 – evaluate some of the ways of promoting sustainable development and assess the effectiveness in selected places and regions of the world.

HE3.03 – assess the contribution of selected government policies to sustainable resource development in Canada.

GC1.05 – identify individuals who have made a significant contribution to addressing global issues and evaluate their impacts.

UC3.06 – evaluate the effectiveness of an international strategy and agreement that has been designed to protect the global commons or address global issues.

UC3.02 – evaluate and communicate the perspectives and arguments of various stakeholders involved in a geographic issue.

Prior Knowledge and Skills

- Knowledge of global environmental trends (climate change, biodiversity, pollution, desertification, deforestation) and a familiarisation with the international agreements that address these trends (Kyoto protocol, the Convention on Biodiversity, etc).
- An understanding of human development trends and how they influence geographic issues (birth/death rates, GDP, child mortality, emissions, etc).
- Statistical Analysis, GIS basic knowledge.

Planning Notes

- Photocopy the Ecological Footprint Questionnaire for student's homework (See Worksheet 1).
- Photocopy Worksheets 2-6.
- Photocopy the Sustainable Development Fact Sheet for distribution (See Appendix 1).
- Photocopy the Sustainable Development Timeline (See Appendix 2).
- Photocopy the Sustainable Development Criteria (See Appendix 3).
- Photocopy the Expanded Version of the Sustainable Development Timeline (See Appendix 4).
- Photocopy the Sustainable Development Objectives for Canada (See Appendix 7).
- Review Suggested Homework 1 and decide if you will assign it.
- Review the Glossary of terms and be prepared to define any of these terms comfortably for the class.

Teaching/Learning Strategies

Most of the learning strategies are described in each activity, but here are some of the overall strategies for the unit.

- Introduce students to the concept of sustainable development. Students will have to demonstrate how our values influence our perceptions on the environment.
- Students explore their behaviours and attitudes towards adopting practices that are more sustainable in Activity 2.
- Introduce students to the concept of the ecological footprint.
- Examine the historical path of sustainable development by identifying international policies and commitments and the impact of their implementation on local, national, and global issues. Students will achieve this through Activity 4.
- Identify individuals who have made a significant contribution to sustainable development and evaluate their impacts in their poster.
- Introduce the students to the new vocabulary and the new concepts listed in each activity. Refer them to the glossary.
- Refer students to the accompanying material, the supplementary sheets, and Resources section of this module.
- Assign students the suggested homework and encourage students to analyse more in-depth the information provided and to research the websites given for this unit.
- Monitor the group work and the simulation games so students can gain maximum benefits from those activities.

Accommodations

Frequent monitoring and assistance should be provided. Accommodations that are normally provided in the regular classroom for students with special needs should be provided. Here are some strategies for helping them.

- Information should be presented to as many modalities as possible through oral presentations, board notes, diagrams, class discussions, activity-based learning.
- Instructions should be clear, concise, to the point, verbally rehearsed by the student, properly sequenced, reviewed after a time lapse, written down on the board or in a notebook.
- Assignments should be outlined in writing, written in the students' homework/assignment book, written (and left) on the blackboard, broken down by step with the teacher, rehearsed / modelled, and scheduled step by step.
- Questioning should be clear, concise, allowing for appropriate response time. Students need time to organise their thoughts, match them with the appropriate vocabulary, put them in correct syntax, sequence the ideas and finally express them with good speech.
- Encourage students to question for clarification and additional information. Learn to speak at a slower rate of speech. Avoid figurative language unless it has been specifically pre-taught. Encourage and evaluate oral responses and class participation. Allow extra time (e.g. homework) for students to complete assignments that others students may finish in class time.